

DSHBeauties - Research Memo 2

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Research Or Analytical Question

What is the distribution of public transportation use frequency among participants?

Answer, Response, + Summary of Results

Among 252 participants with complete responses to Q26, public transportation use frequency varied across participants, with responses distributed relatively evenly across the five categories. As shown in Figure 1, no single response category dominated the distribution.

The largest group reported using public transportation **8 or more days a month** ($n = 56$, 22.2%), followed by **2–4 days a month** ($n = 54$, 21.4%). The smallest group reported using public transportation **0–1 days a month** ($n = 40$, 15.9%), while 51 participants (20.2%) reported **never using** public transportation.

Overall, the distribution suggests that Q26 captures variation in transportation use and provides a useful basis for examining factors associated with transportation behavior in future analyses.

```
# Define the correct ordinal order
q26_order = [
    "Never",
    "0-1 days a month",
    "2-4 days a month",
    "4-8 days a month",
    "8 or more days a month"
]

# Frequency counts
q26_counts = (
    df["Q26"]
    .value_counts()
    .reindex(q26_order)
)

# Percentages
q26_percent = q26_counts / q26_counts.sum() * 100

# Create dataframe
plot_df = pd.DataFrame({
    "Usage Frequency": q26_order,
    "Count": q26_counts.values,
    "Percentage": q26_percent.values
})

# Plot
sns.set_theme(style="white")
```

```

plt.figure(figsize=(6.5, 2.8))

ax = sns.barplot(
    data=plot_df,
    y="Usage Frequency",
    x="Percentage",
    color="#4C72B0",
    width=0.6
)

# Add labels
label_x = max(plot_df["Percentage"]) + 1

for i, row in plot_df.iterrows():
    ax.text(
        label_x,
        i,
        f'{row["Percentage"]:.1f}% (n={row["Count"]})',
        va="center",
        ha="left",
        fontsize=8
    )

# Formatting
plt.xlabel("Participants (%)", fontsize=9)
plt.ylabel("")

plt.yticks(fontsize=9)
plt.xticks(fontsize=9)

plt.xlim(0, 30)

sns.despine()

plt.tight_layout()

plt.show()

```

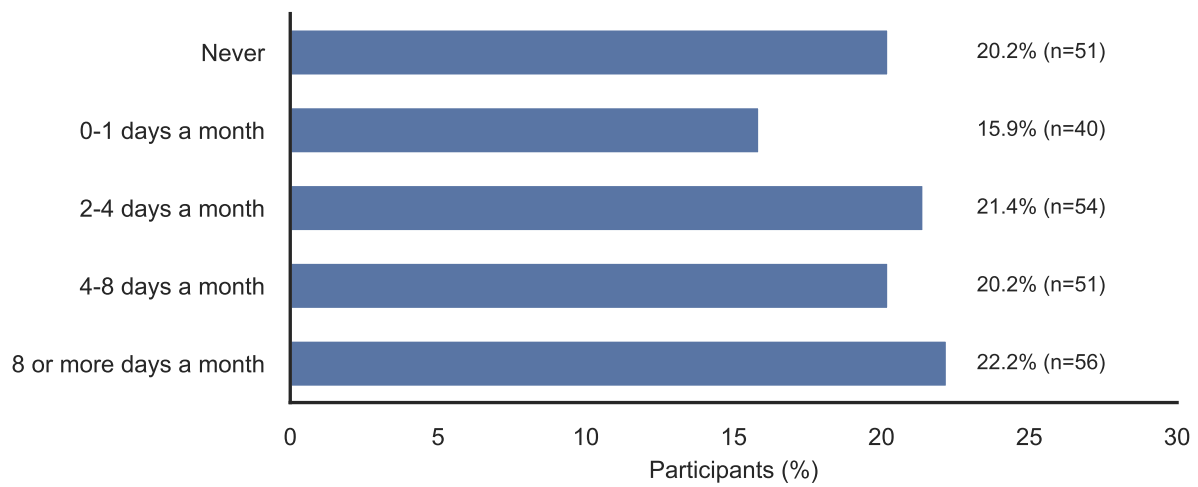


Figure 1: Distribution of Public Transportation Use Among Participants

What questions or uncertainties remain?

- How should public transportation use frequency categories be appropriately coded for future statistical analysis?
- What factors may explain differences in public transportation use frequency among participants?